Tutorial

A healthcare system wants to predict whether a patient has diabetics using clinical distribute

Input features:

- Age
- BMI
- Blood glucose level
- Blood pressure

Target variable : Diabetics (1 = Yes, 0 = NO)

Age	BMI	Glucose	Blood pressure	Diabetics
45	28	160	85	1
30	22	110	70	0
50	31	180	90	1

Questions:

1. Identify the classification type of this problm.
2. write the logistic regression hypothesis function
3. Given predicted probabilities, compute the log loss value for 3 iterations.

Ans:

1. Target value is diabatics → 0, 1 only 2 classes so their is a binary classification cus output

$\sigma(z) = 0.8909$

$h_2 = 1.7 \quad \sigma(z) = 0.8455$

$h_3 = 2.6 \quad \sigma(z) = 0.9308$

$h_4 = 3.2 \quad \sigma(z) = 0.9608$

$h_5 = 2.8 \quad \sigma(z) = 0.9426$

O/P layer calculation:

$$z = 0.2(0.8909) + 0.1(0.8455) + 0.3(0.930) + 0.2(0.960) + 0.4(0.94) + 0.1$$

$z = 1.20968$

$$y_p = \frac{1}{1+e^{-1.2096}} = 0.77 \cong 1$$

$y_a = 1$

$\therefore y_p \approx y \Rightarrow$ Yes//

$\therefore$ user will exercize as $y \cong y_p$ (yes = 1)

Activation function

Sigmoid function $= \frac{1}{1+e^{-z}}$

$\sigma(z)$

learning parameters $\Rightarrow \eta = 0.1$

Initial weights

h_1 $w_1 = 0.2$ $w_2 = 0.3$ $w_3 = 0.1$

h_2 $w_4 = 0.1$ $w_5 = 0.4$ $w_6 = 0.2$

h_3 $w_7 = 0.3$ $w_8 = 0.2$ $w_9 = 0.1$

h_4 $w_{10} = 0.4$ $w_{11} = 0.1$ $w_{12} = 0.2$

h_5 $w_{13} = 0.3$ $w_{14} = 0.3$ $w_{15} = 0.1$

O/p layer weights

$v_1 = 0.2$ $v_2 = 0.1$ $v_3 = 0.2$ $v_4 = 0.2$ $v_5 = 0.4$

$b = 0.1$

Training ex:

$x_1 = 7$ $x_2 = 2$ $x_3 = 1$

$h_1 = w_1 x_1 + w_2 x_2 + w_3 x_3 = 2.1$

has only 2 labels i.e Yes or NO [0 or 1]

2. $z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4$

$z = w_0 + w_1(\text{Age}) + w_2(\text{BMI}) + w_3(\text{glucose}) + w_4(\text{BP})$

sigmoid function

$$\sigma(z) = \frac{1}{1+e^{-z}} = \sigma(z) = \frac{1}{1+e^{-(w_0 + w_1 x_1 + w_1 x_2 + w_3 x_3 + w_4 x_4)}}$$

(3) loss function

$$J = -[y \log(\hat{y}) + (1-y)\log(1-\hat{y})]$$

→ Normalization is a process of scaling numerical features to common range usually [0,1] or [-1,1]

$$x = \frac{x - x_{min}}{x_{max} - x_{min}}$$

After normalisation

age	BMI	glucose	BP	y
0.8	0.73	0.76	0.72	1
0.2	0.18	0.18	0.09	0
1	1	1	1	

$w_0, w_1, w_2, w_3, w_4, w_5 = 0.01$

$\alpha = 0.1$

gradient descent

$w_{new} = w_{old} - \text{learning rate } (\text{error})\, x_i$

$w_0 = 0.01 - (0.1)(-0.49) = 0.059$

$w_1 = 0.01 (0.1) \cdot (-0.495)(0.85) = 0.05$

$w_2 = 0.01 - (0.1)(-0.49)(0.73) = 0.0484$

$w_3 = 0.01 - (0.1)(-0.49)(0.76) = 0.0472$

$w_4 = 0.01 - (0.1)(-0.49)(0.72) = 0.045$

Sample ②

$z = 0.059 + 0.05(0.2) + (0.05)(0.18) + (0.04)(0.15) + (0.04)(0.09) = 0.088$

$\hat{y} = \frac{1}{1+e^{-z}} = 0.0706$

loss function

$J = -[0 + (1-0)\log(1-0.706)]$

$= -[1 \log(1-0.706)]$

$= 0.531$

error $(\hat{y} - y) = (0.706)$

gradient descent

$w_0 = 0.059 - (0.1)(0.706)(1) = -0.0116$

$w_1 = 0.05 - (0.1)(0.706)(0.2) = 0.035$

first sample :

$1 = 0.01 + 0.01(0.8) + 0.01(0.73) + (0.01)(0.76) + 0.01(0.77)$

$1 = 0.0406$

$\hat{y}_1 = \frac{1}{1+e^{-z}} \Rightarrow \frac{1}{1+e^{-(0.0406)}}$ $\hat{y}_1 = 0.510$

log loss $(J) = -\log(\hat{y})$ when $y=1$
$y=0$

$J = -\log(1-\hat{y})$

$\therefore J_1 = -\log(\hat{y})$

$J_1 = 0.292$

$E_1 \Rightarrow \hat{y} - y_1 = 0.510 - 1 \Rightarrow -0.49$

$z_2 = 0.01 + 0.01(0.2) + (0.01)(0.18) + (0.01)(0.18) + 0.01(0.09)$

$z_2 = 0.0165$ $z_3 = 0.05$

gradient descent

$w_{new} = w_{old} - \text{learning rate} \sum (\text{error}) \cdot x$

$= 0.01 - 0.1(-0.49) \Rightarrow 0.0149$

x_1 Study hrs	x_2 Attendance (%)	x_3 Assignment score	x_4 practice tests	y final examscore	$\hat{y}$	error $y-\hat{y}$	w_1	w_{new}	$\hat{y}$
12	85	78	3	80	0	80	$w_1=0$	3.95	260506.45
15	90	82	4	85	0	85	$w_2=0$	248.5	275554.65
10	75	70	2	72	0	72	$w_3=0$	1639.25	231450.7
18	95	88	5	90	0	90	$w_4=0$	1512.85	293632.75
8	70	65	1	68	0	68	$w_5=0$	59.25	215133.95
				395		395			